



AgriAgent

Biopesticides that survive the sun

Autonomous SCADA for on-demand biopesticide formulation — Perception → Formulation → Safety →
Actuator

Hackathon pitch · 3 minutes



THE PROBLEM

Sunlight destroys the cure

8.8 h

Bright dry midday

UV 9.2 · 38 °C · 32 % RH

37 h

Overcast morning

UV 2.1 · 24 °C · 75 % RH

72 h

Night

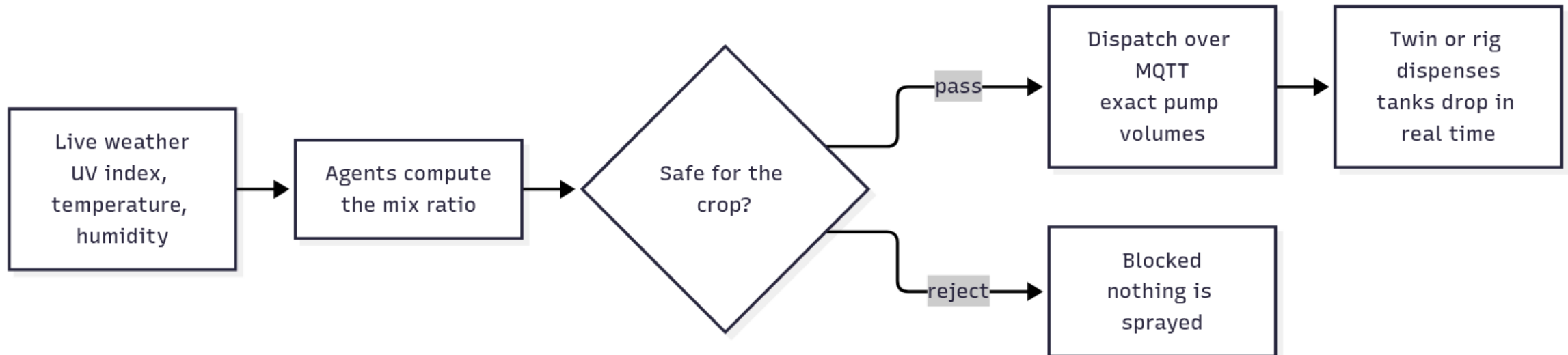
UV 0 · 18 °C · 85 % RH

Bt half-life on the leaf — an 8× swing in a single day. A tank mixed at dawn is largely spent by afternoon, so farmers spray more often: higher cost, and the biological advantage is lost anyway.



SCOPE

Not a platform. One decision.



Minimum data

UV index, temperature, humidity — pulled from live weather. Farmers never hand-enter data.

Actionable output

One MQTT payload: recipe → safety gate → exact pump volumes → the twin dispenses them.

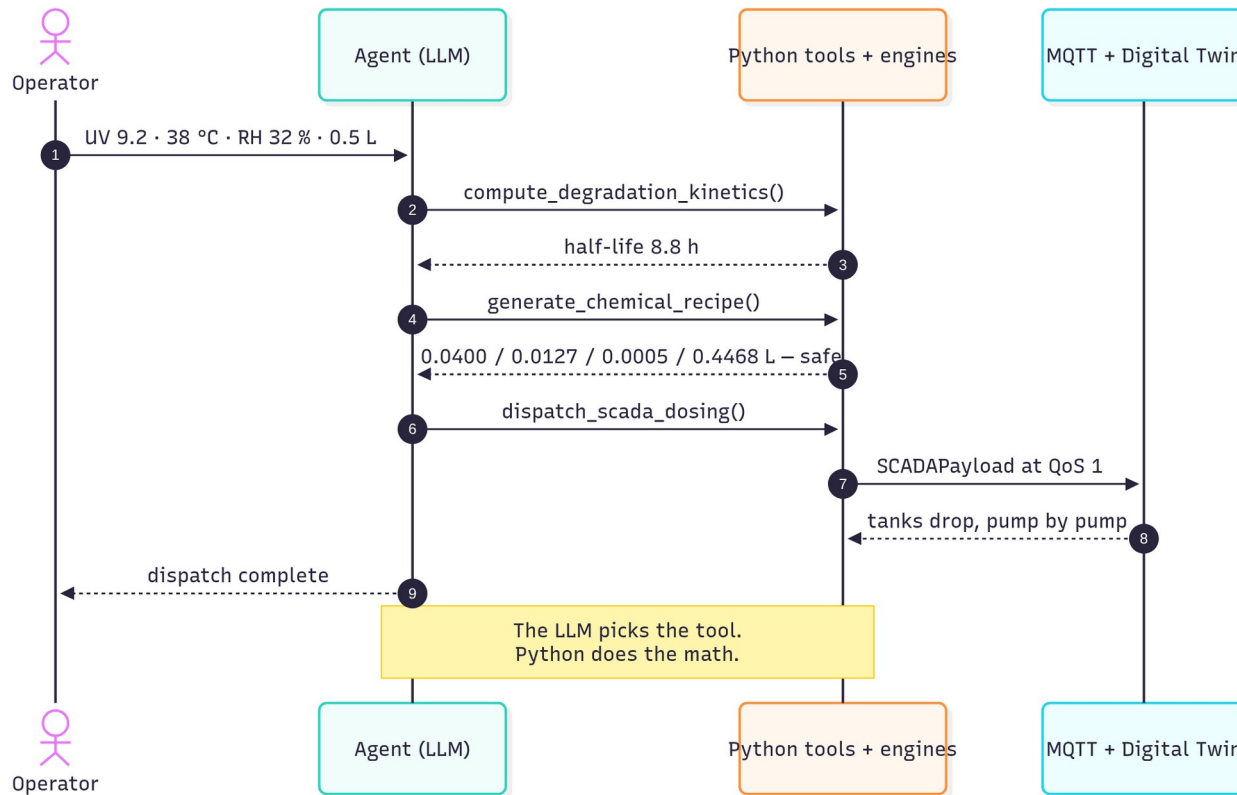
Validated savings

Kinetics models tie the dynamic mix to the 90 % synthetic-volume-reduction target.



ARCHITECTURE

The LLM picks the tool. Python does the math.



- **Perception**
reads telemetry, computes the degradation half-life
- **Formulation**
exact litre setpoints for all four channels
- **Safety**
machine-readable rejections — nothing unsafe is published
- **SCADA**
pump runtimes, MQTT dispatch at QoS 1



PROOF

The chemistry is not hallucinated

Confirmed kinetics

$$k = k_0 \cdot (1 + \alpha \cdot UV) \cdot e^{-(E_a/R)(1/T - 1/T_0)}$$

UV photolysis × Arrhenius thermal · $k_0 = \ln(2)/48 \text{ h}^{-1}$

$$\text{lignin} = \min(3.0 \%, 0.25 \% + 0.25 \% \cdot UV)$$

UV stabilizer, capped at the solubility limit

$$\text{surfactant} = \min(0.20 \%, 0.05 \% \cdot (1 + 1.2(1 - RH/100)) \cdot (T/293.15)^{1.5})$$

wetting agent, capped against phytotoxicity

$$\text{runtime} = \text{volume} \div \text{flow}$$

0.01 / 0.01 / 0.01 / 0.05 L/s — never accepted from a caller

```
$ python src/mcp_server_scada.py --selftest
[PASS] k_baseline: got=0.078508
[PASS] half_life: got=8.829
[PASS] bio_l: got=0.04
[PASS] lignin_pct: got=2.55
[PASS] surf_pct_capped: got=0.0993
[PASS] rejects_overcap_surfactant: True
[PASS] pump1_duration: got=4.0
[PASS] all_runtimes_under_30s: True
[PASS] lignin_cap: got=3.0
... 21 checks
SELFTEST PASSED
```



One dispatch, end to end

<<

Telemetry

UV index

9.20

Temperature (°C)

38.00

Relative humidity (%)

32

Batch volume (L)

0.50

Preview only — nothing is dosed until you dispatch

Dispatch

Dispatch now

DISPATCH SUCCESS to agri/actuator/zone_north/dosing_dispatch

AgriAgent SCADA

Live biopesticide formulation — Perception → Formulation → Safety → Actuator

Degradation rate k

0.0785 /hr

Half-life

8.8 h

Viability after 1h

92.4%

Recipe

Biopesticide

0.0400 L

UV stabilizer

0.0127 L

Surfactant

0.0005 L

Carrier water

0.4468 L

Batch total

0.50 L

✔ Safety check passed — recipe approved.

📄 Dispatch payload

Digital twin

Tank levels

Biopesticide

4.920 L

UV stabilizer

1.974 L

Surfactant

0.999 L

Carrier water

49.106 L

Latest batch

0.500 L

Total dispensed

1.000 L

Zone zone_north · last updated 1791024827.1650212

Deploy ⋮

The recipe

bio 0.0400 L · stabilizer 0.01275 L ·
surfactant 0.0005 L · water 0.44675 L

The runtimes

4.000 s · 1.275 s · 0.050 s · 8.935 s —
computed from the calibrated flow table

The feedback

tank state published after every pump, then
DISPATCH_COMPLETE



Refusing is the feature

10 L batch requested

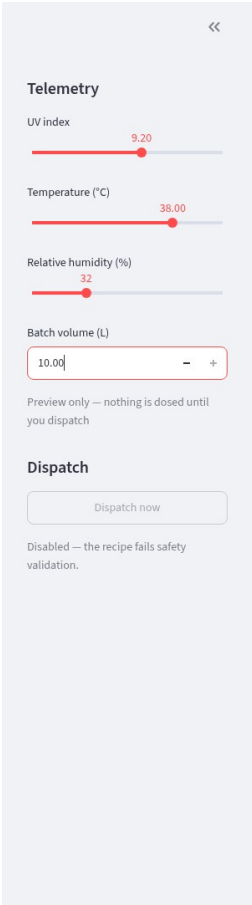
pump 1 runtime 80.0 s, pump 4 runtime 178.7 s — both breach the 30 s thermal cap

Blocked before you can click

the dispatch button disables itself while the recipe fails validation

Nothing is published

no MQTT payload, tanks do not move — the rejection is machine-readable



AgriAgent SCADA

Live biopesticide formulation — Perception → Formulation → Safety → Actuator

Degradation rate k 0.0785 /hr	Half-life 8.8 h	Viability after 1h 92.4%
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Recipe

Biopesticide 0.8000 L	UV stabilizer 0.2550 L	Surfactant 0.0099 L	Carrier water 8.9351 L	Batch total 10.00 L
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Safety check FAILED — pump 1 runtime 80.0s exceeds 30s threshold; pump 4 runtime 178.7s exceeds 30s threshold

Dispatch payload

Digital twin

Tank levels

Biopesticide 4.920 L	UV stabilizer 1.974 L	Surfactant 0.999 L	Carrier water 49.106 L
Latest batch 0.500 L	Total dispensed 1.000 L		

Zone zone_north - last updated 1791024827.1650212

Deploy



IMPACT

One decision, made correctly, every cycle

90 %

synthetic-volume reduction target

PKR 45,000

per-acre saving (cost-model estimate)

7–14×

half-life extension, lignin encapsulation
(literature)

Honest limits — targets, not measurements

- The 90 % and PKR figures are literature-backed targets and cost-model estimates — not results this build measured.
- The physical ESP32 rig is not built: the digital twin is the demo target, and firmware inherits the same MQTT contract.
- The twin simulates timed pump actuation and tank depletion — not fluid mechanics.